



Task 1

Text Watermark Localization



Language Models: Autoregressive Predictions

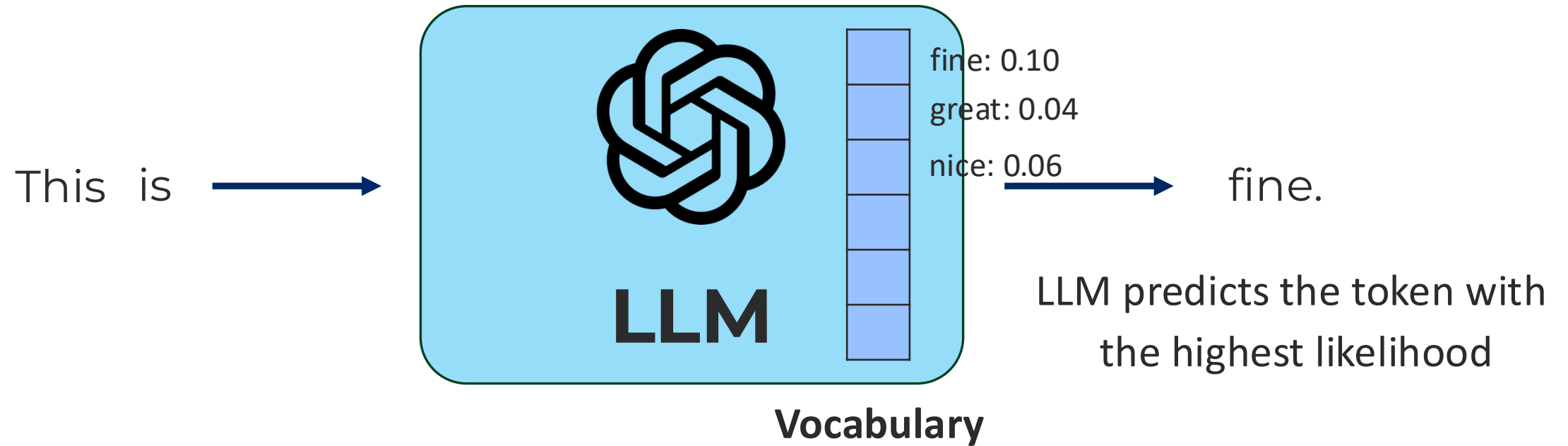
This



is



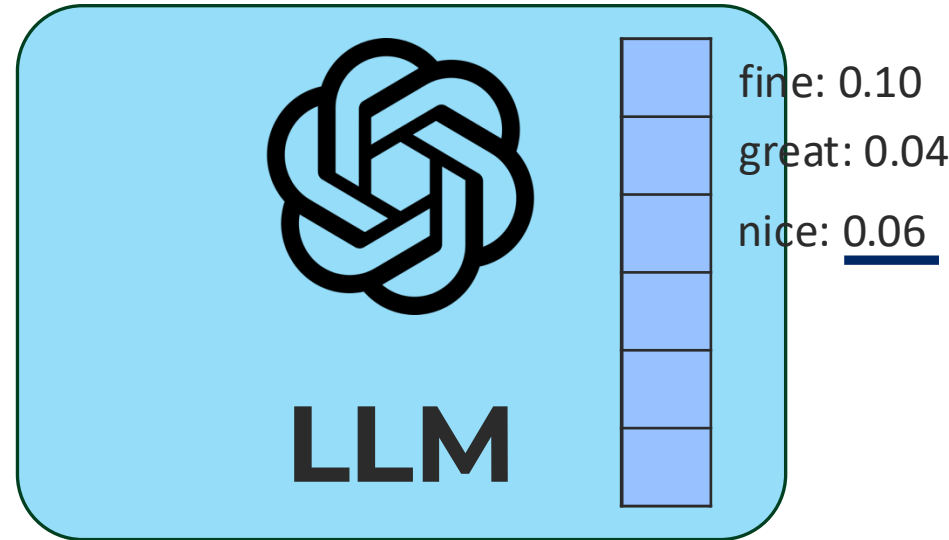
Language Models: Autoregressive Predictions





Language Model Watermarking: Red-Green List

This is



No watermark
Extremely efficient on average term lengths and word frequencies on synthetic, microamount text (as little as 25 words)
Very small and low-resource key/hash (e.g., 140 bits per key is sufficient for 99.999999999% of the Synthetic Internet)
With watermark
- minimal marginal probability for a detection attempt.
- Good speech frequency and energy rate reduction.
- messages indiscernible to humans.
- easy for humans to verify.

Intuition: Divide the vocabulary in “red” and “green” tokens at random

Normal (non-watermarked) text: Red and green tokens are equally likely

Watermarked text: Green tokens are more likely



Language Model Watermarking: Red-Green List

Algorithm 1 Text Generation with Hard Red List

Input: prompt, $s^{(-N_p)} \dots s^{(-1)}$

for $t = 0, 1, \dots$ **do**

1. Apply the language model to prior tokens $s^{(-N_p)} \dots s^{(t-1)}$ to get a probability vector $p^{(t)}$ over the vocabulary.
2. Compute a hash of token $s^{(t-1)}$, and use it to seed a random number generator.
3. Using this seed, randomly partition the vocabulary into a “green list” G and a “red list” R of equal size.
4. Sample $s^{(t)}$ from G , never generating any token in the red list.

end for

Get the probability vector

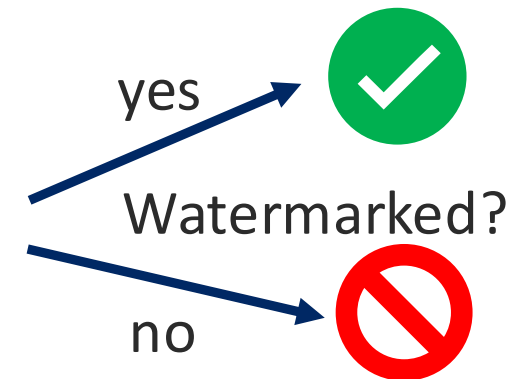
fine: 0.10
great: 0.04
nice: 0.06

Use previous token to generate random number

Use the seed to split vocab into red & green lists

Sample tokens from green list

Perform statistical testing: are green tokens in the text significantly more likely than random chance?





Task 1: Text Watermark Localization

Identify **Watermarked** Text Regions:

Stopped 30 of 34 shots in the Blue
Jackets' 6-2 win over the Canadiens
on Monday. The loss dropped him to
30-22-1 on the season with a 2.76
GAA and .906 The victory pushed him
to 31-21-1 on the season with a 2.72
GAA and .908 save percentage.
Mar.
10 11:49 AM PT12:49 PM
[...continues]

loss dropped ... – **non-watermarked** text

The victory ... – **watermarked** text



Task 1: Text Watermark Localization

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[...continues]

loss dropped ... – non-watermarked text

The victory ... – watermarked text

Watermarking schemes fully provided:

1. TextSeal
2. Gumbel-Max
3. Unigram
4. KGW (red-green watermark)

Non-watermarked text:

1. Human-created
2. Non-watermark generation
3. Different watermarks than the above



Task 1: Score for the Text Watermark Localization

Identify **Watermarked** Text Regions:

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GAA and .908 save percentage.\nMar.
10 11:49 AM PT12:49 PM
[...continues]

Assign a continuous **Score**
in range [0,1] to each token:

Stopped: 0.8321
30: 0.722

....

Score ≥ 0.5 : token is from a
watermarked text region

Score < 0.5 : token is clean



Task 1: Score for the Text Watermark Localization

Identify **Watermarked** Text Regions:

Assign a continuous **Score** in range $[0,1]$ to each token:

Score ≥ 0.5 : token is from a watermarked text region

Score < 0.5 : token is clean

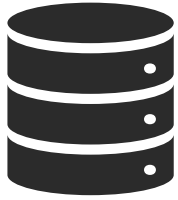
The final score across all the tokens:

TPR @ 0.1% FPR



Task 1: Text Watermark Localization

Data:



1500 text documents:

Training set: 90 documents

Validation set: 90 documents

Test set: 1320 documents



Task 1: Text Watermark Localization

Data:



1500 text documents:

Training set: 90 documents

Validation set: 90 documents

Test set: 1320 documents

Training/Validation record:

```
{  
  "document_id": "train_1",  
  "text": "Paul's statement",  
  "token_ids": [25300, 594, 5114],  
  "token_pieces": ["Paul", "'s",  
    "statement"],  
  "labels": [0, 0, 1]  
}
```



Task 1: Text Watermark Localization

Data:



1500 text documents:

Training set: 90 documents

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Test set: 1320 documents

Test record:

```
{  
  "document_id": "train_1",  
  "text": "Paul's statement",  
  "token_ids": [25300, 594, 5114],  
  "token_pieces": ["Paul", "'s",  
    "statement"]  
}
```



Task 1: Text Watermark Localization

Submission format (example):

```
{"document_id": "1", "scores": [0.02, 0.04, 0.91, 0.87, ...]}  
{"document_id": "2", "scores": [0.12, 0.16, 0.21, ...]}  
...
```



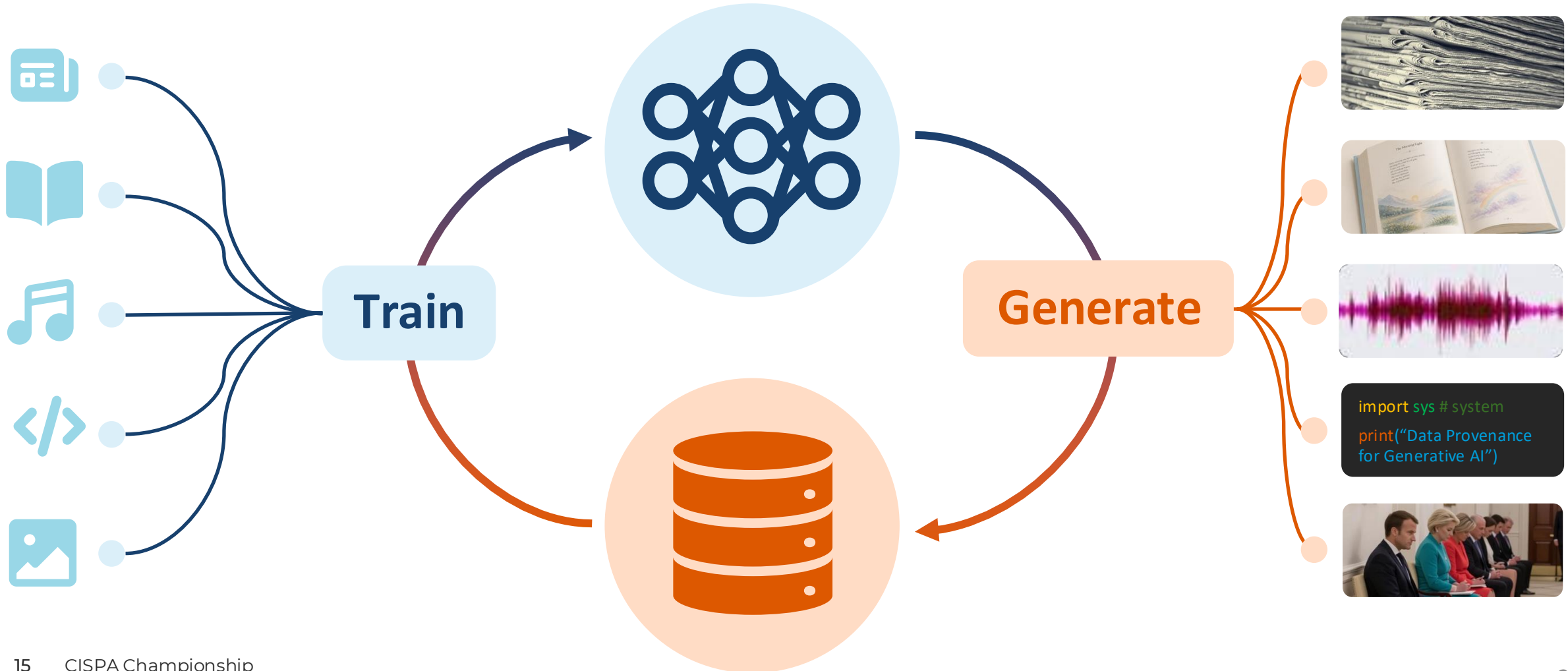
Task 2

Member vs Generated Inference (MGI)



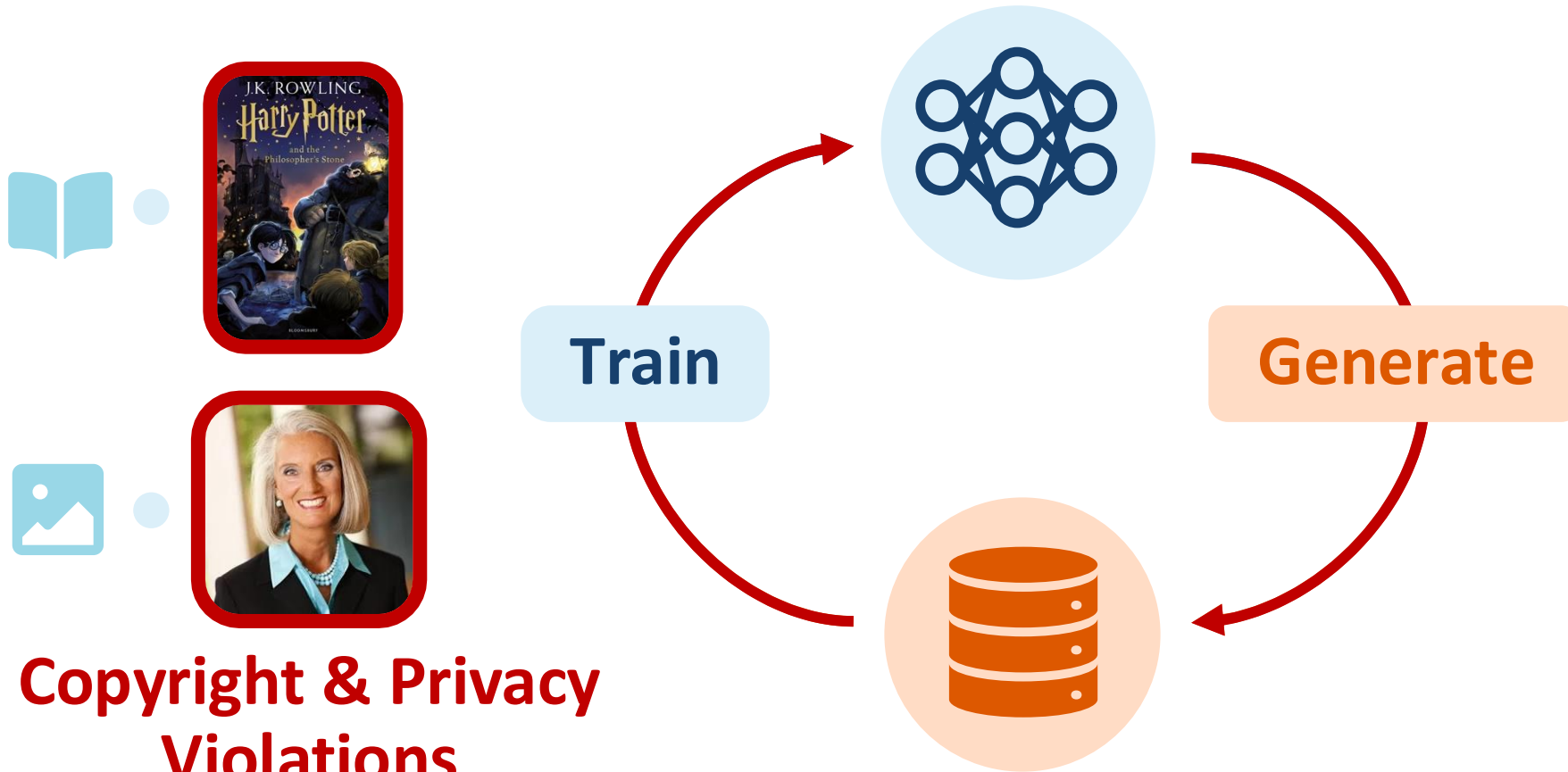
Our Large Data Leads to Impressive Capabilities

Generative Models:  GPT  Gemini ...  LLaMA





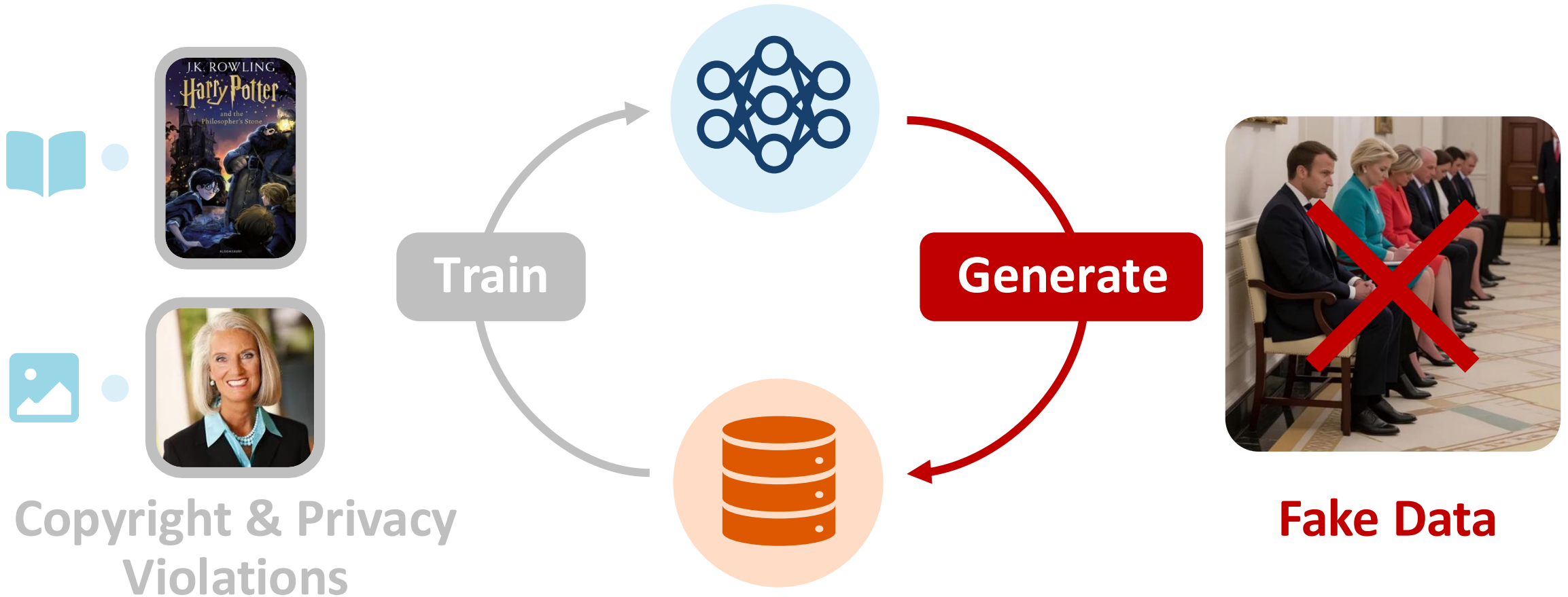
Generative Models Memorize Training Data



> 100 lawsuits, 100 bn €

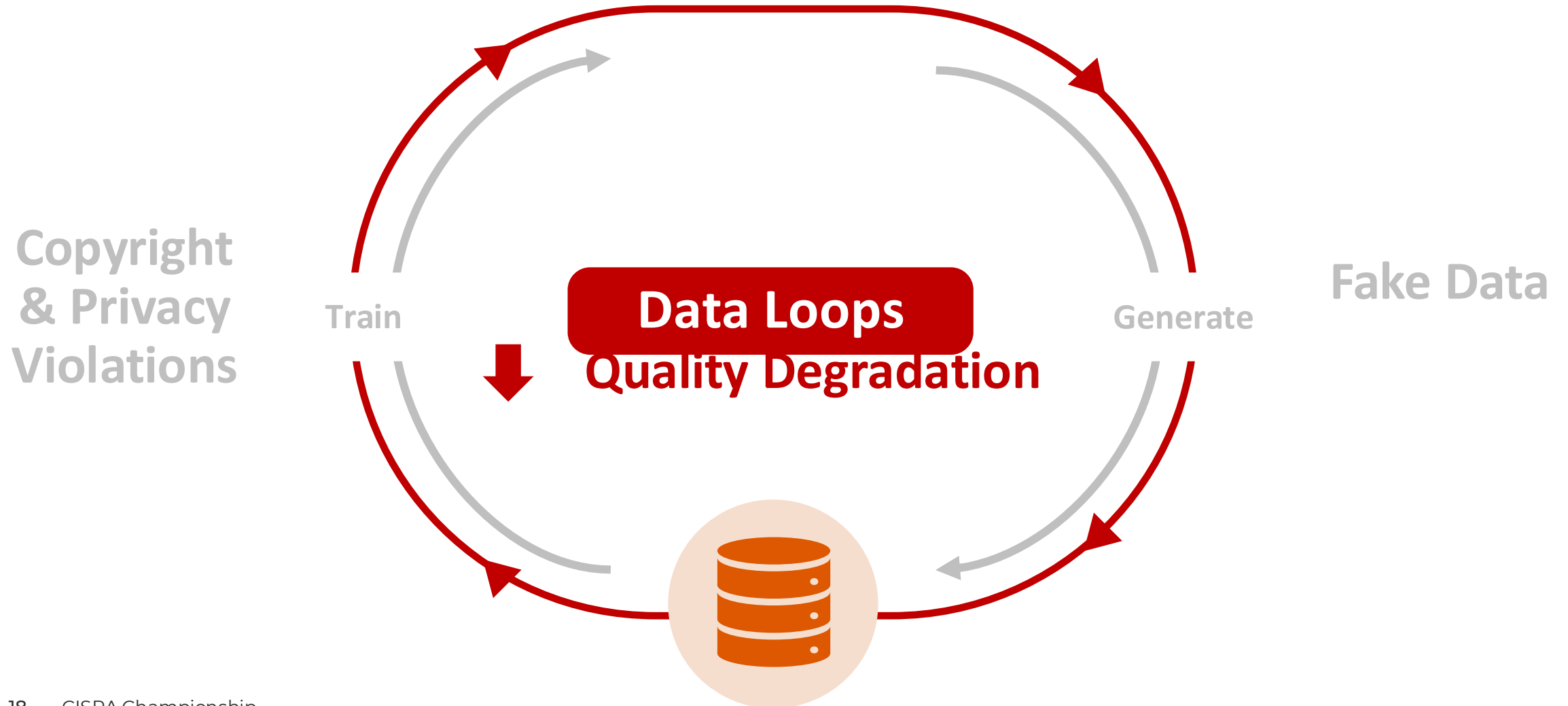


Generative Models Produce Misinformation



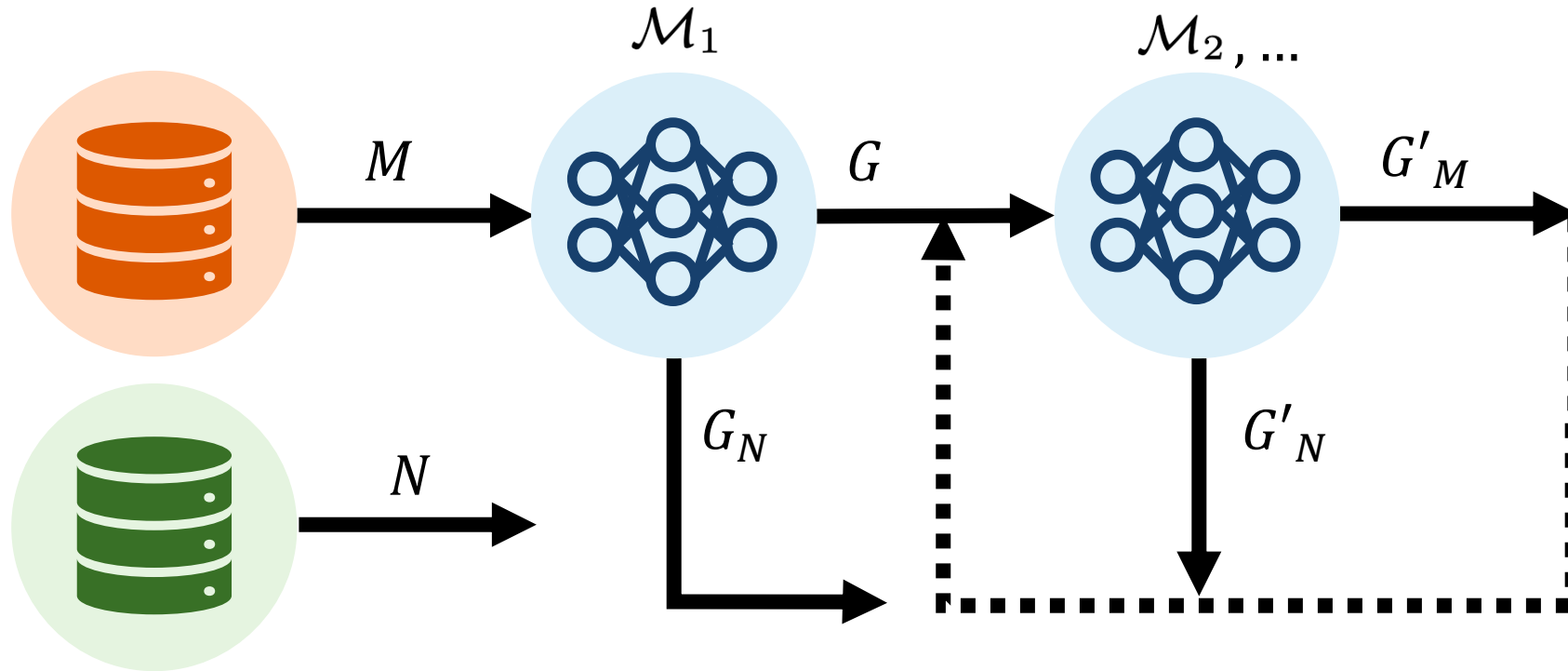


Generative Models Create Data Loops





Task 2: Member vs Generated Inference (MGI)



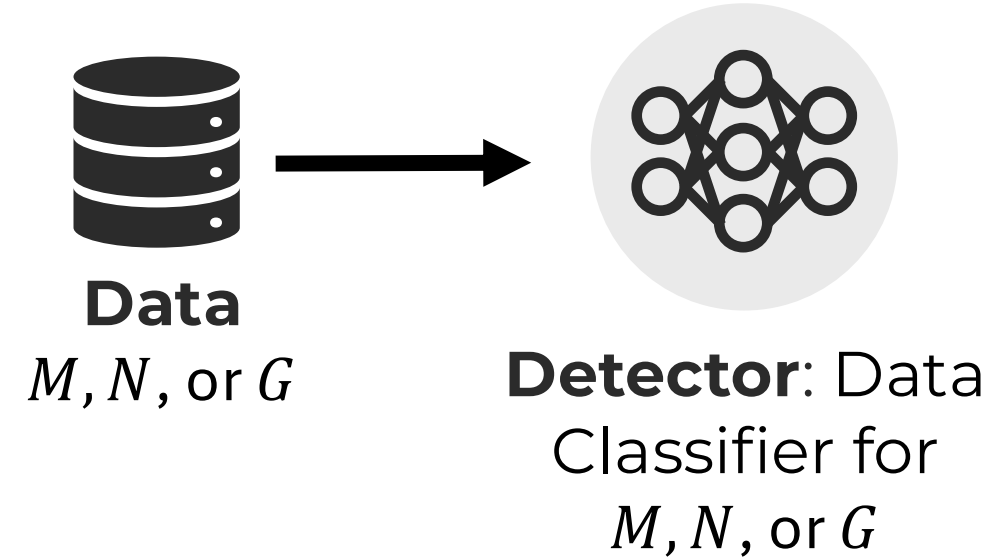
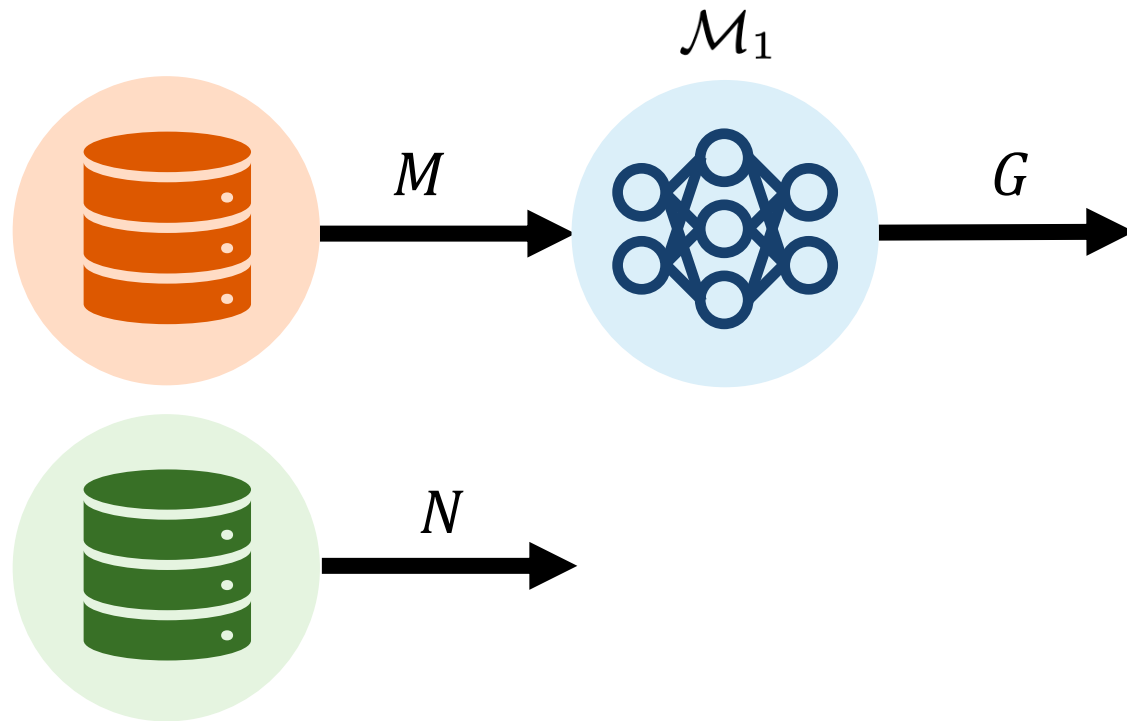
M – Member training samples for model $\mathcal{M}_1$

N – Non-member data from the same distribution as data M

G – Generated data from model $\mathcal{M}_1$



Task 2: Member vs Generated Inference (MGI)



Fool the Detector to misclassify:

$$M \rightarrow N$$

$$M \rightarrow G$$

$$N \rightarrow M$$

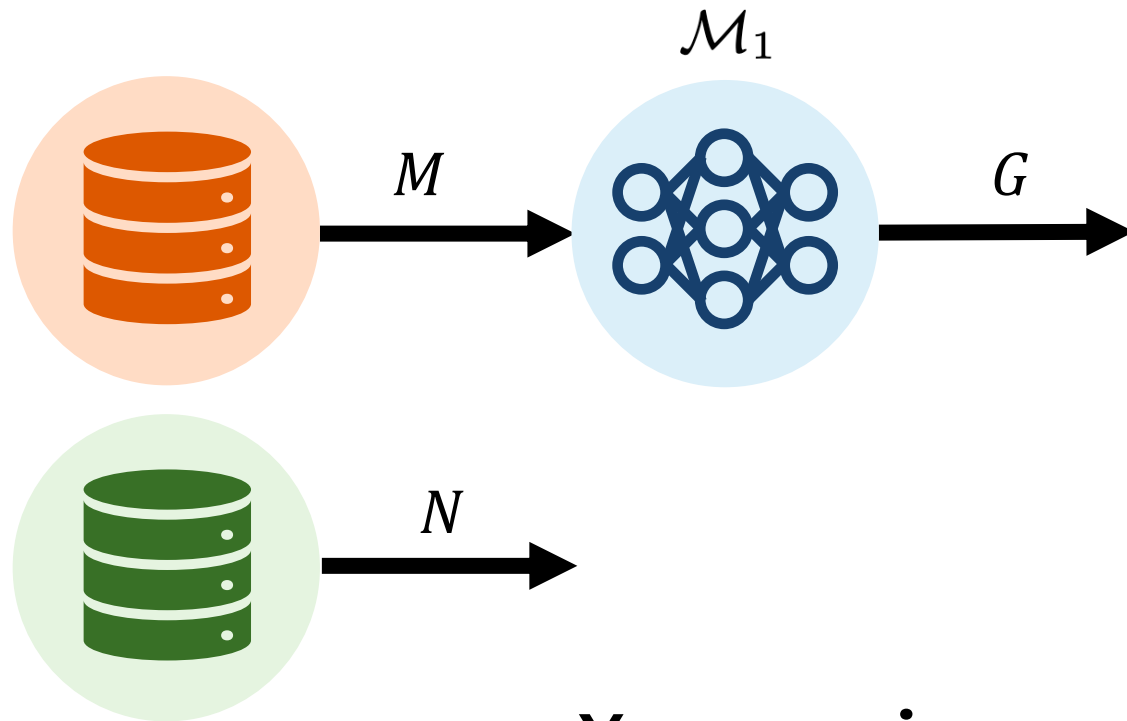
$$N \rightarrow G$$

$$G \rightarrow M$$

$$G \rightarrow N$$



Task 2: Member vs Generated Inference (MGI)

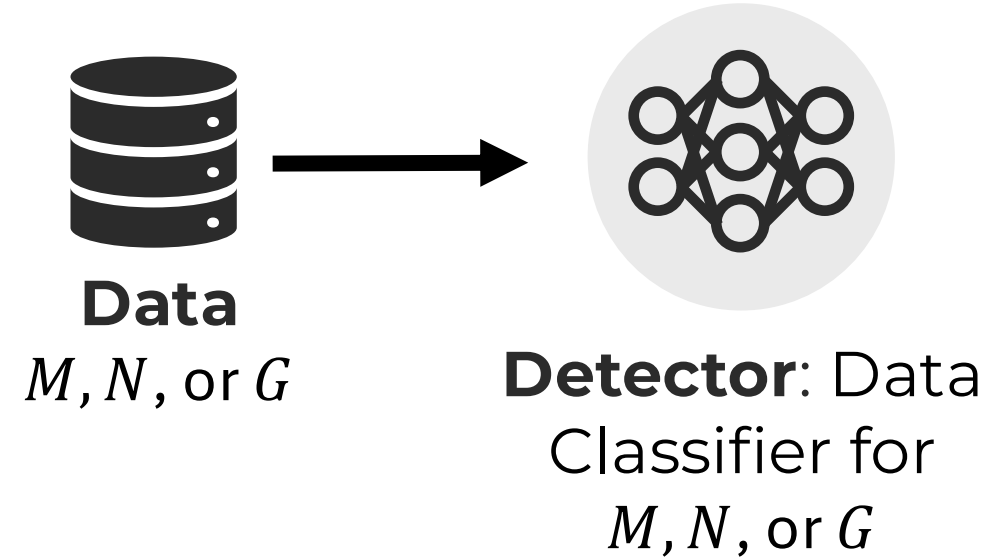


You are given:

M – 300 samples

N – 300 samples

G – 300 samples



Fool the Detector to misclassify:

$M \rightarrow N$

$M \rightarrow G$

$N \rightarrow M$

$N \rightarrow G$

$G \rightarrow M$

$G \rightarrow N$



Task 2: Member vs Generated Inference (MGI)

The submission format:

$M \rightarrow N$ (300 examples starting from the M set)

$M \rightarrow G$ (300 examples starting from the M set)

$N \rightarrow M$ (300 examples starting from the N set)

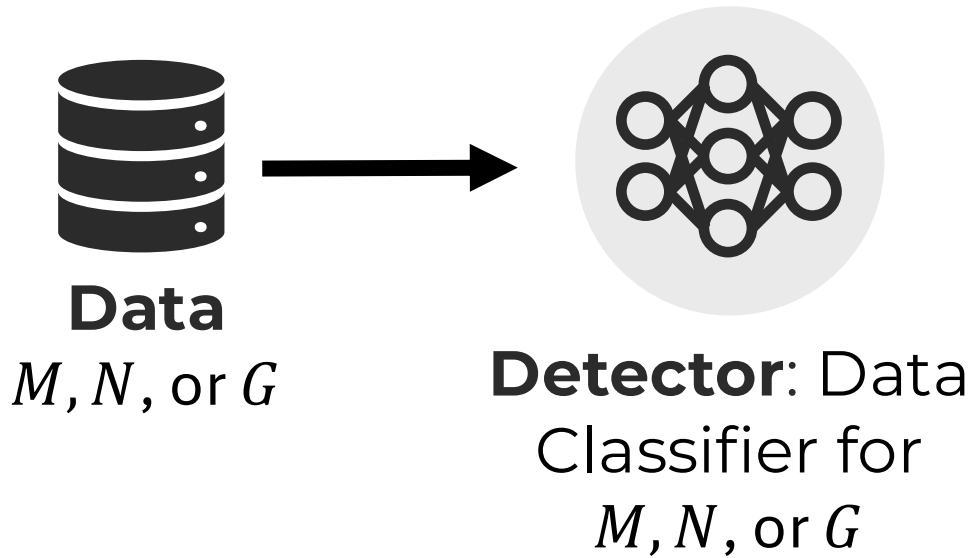
$N \rightarrow G$ (300 examples starting from the N set)

$G \rightarrow M$ (300 examples starting from the G set)

$G \rightarrow N$ (300 examples starting from the G set)



Task 2: Member vs Generated Inference (MGI)



Fool the Detector to misclassify:

$M \rightarrow N$

$M \rightarrow G$

$N \rightarrow M$

$N \rightarrow G$

$G \rightarrow M$

$G \rightarrow N$

Assessment metrics:

1. **DetectScore** = 1 if sample misclassified and 0 otherwise.
2. Image **i**-th Quality: $MSE_i = \frac{1}{m} |\mathbf{X} - \mathbf{X}'|^2$, where **m** is number of pixels, **X**- the initial image, **X'**- your modified image. The lower **MSE**, the better.
3. You have to return **n=1800** images, for which the final **score** $\in [0, 1]$ is:

$$\text{score} = \frac{1}{n} \sum_{i=1}^n \text{DetectScore} \times (1 - MSE_i)$$

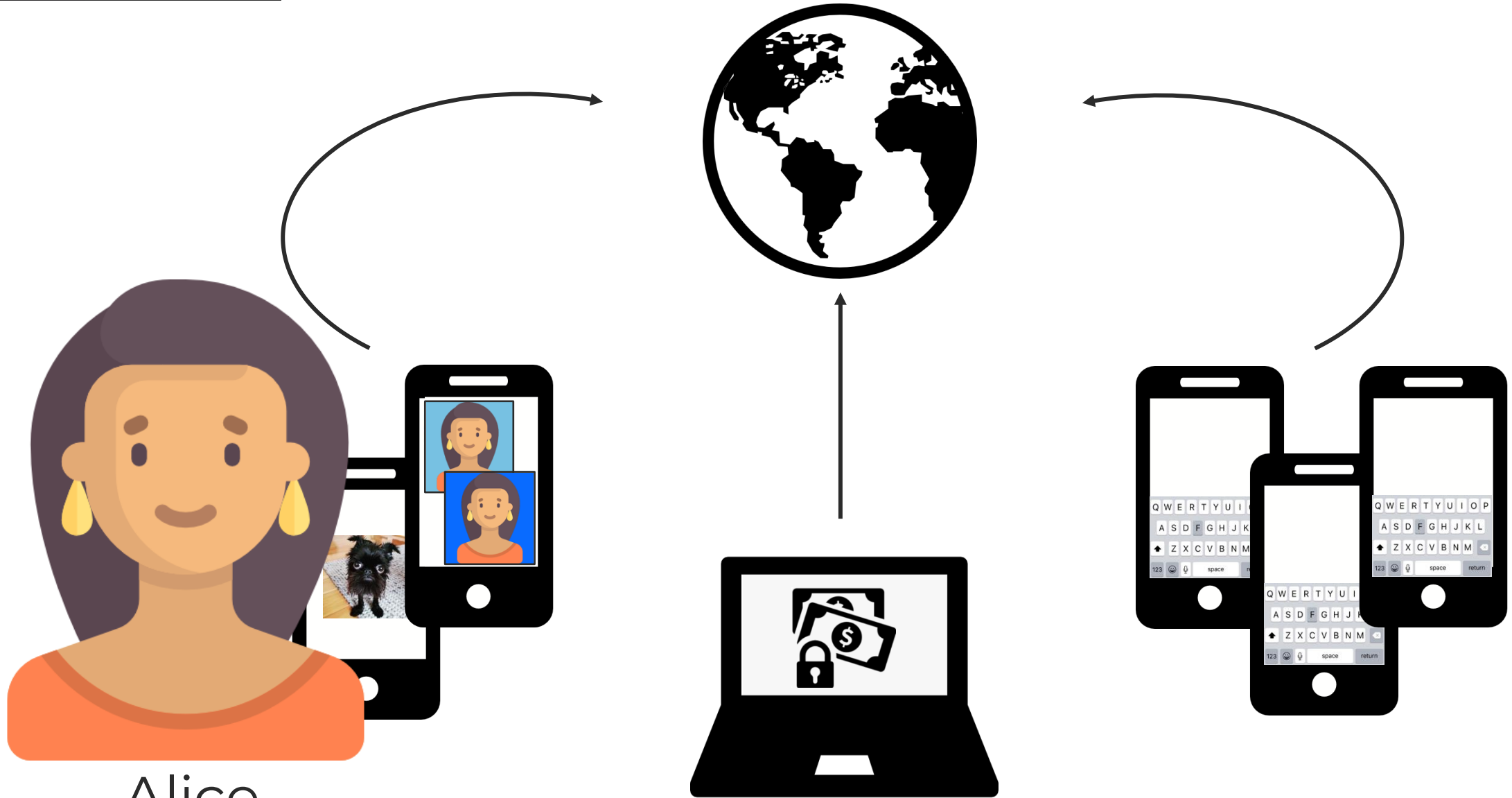


Task 3

Data Reconstruction from Gradients



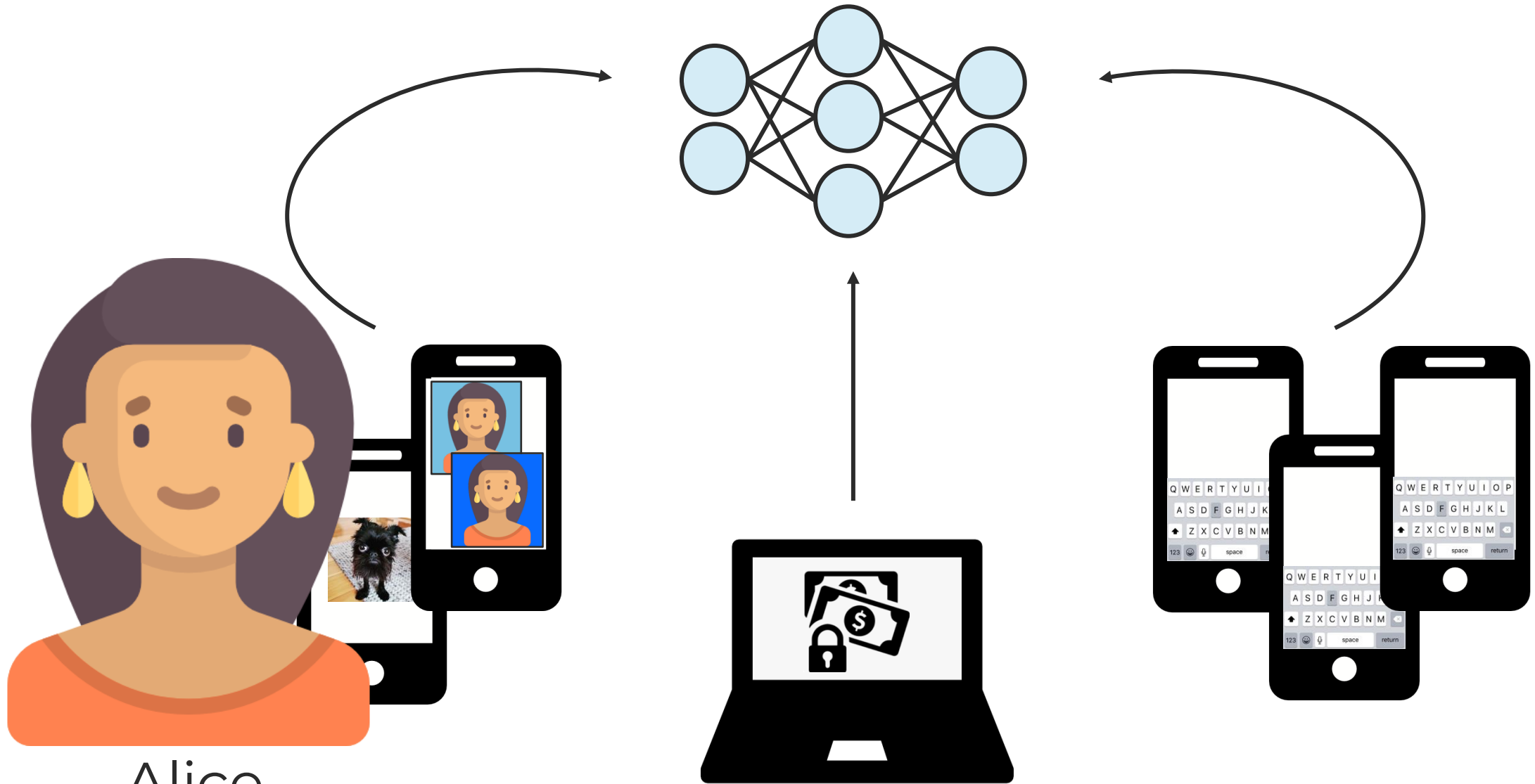
Individuals Generate Sensitive Data



Alice



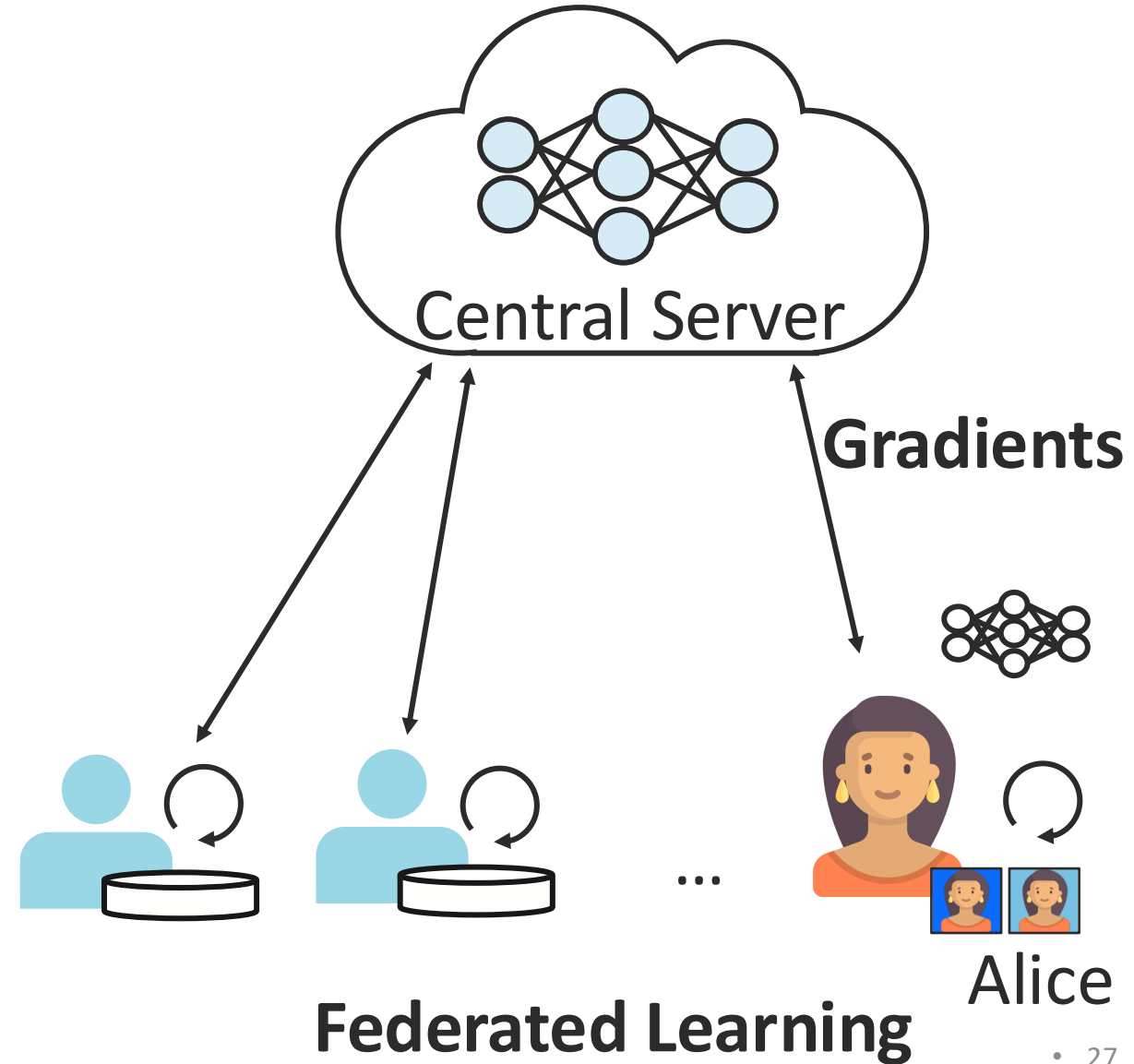
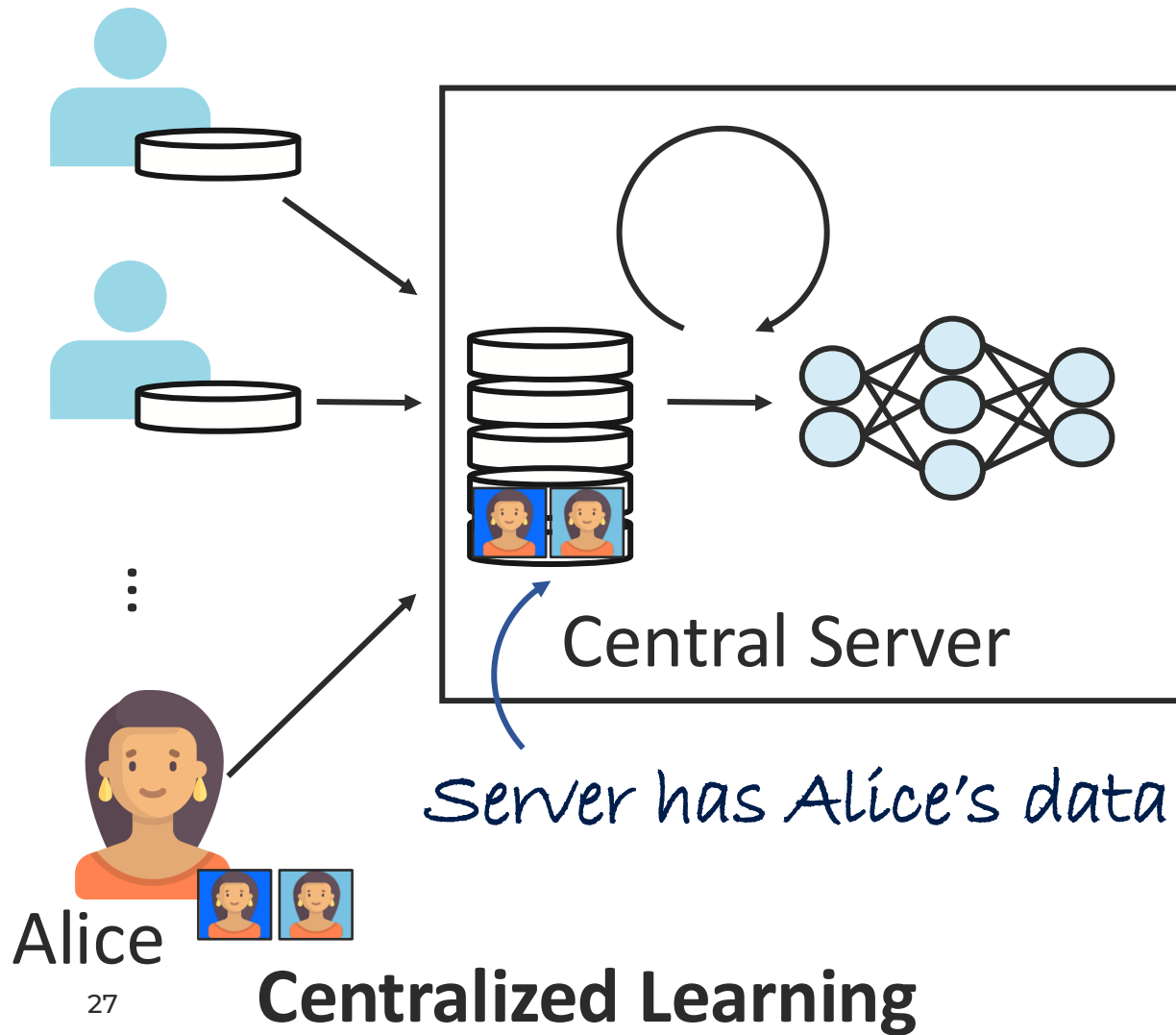
Companies Train Machine Learning Models



Alice

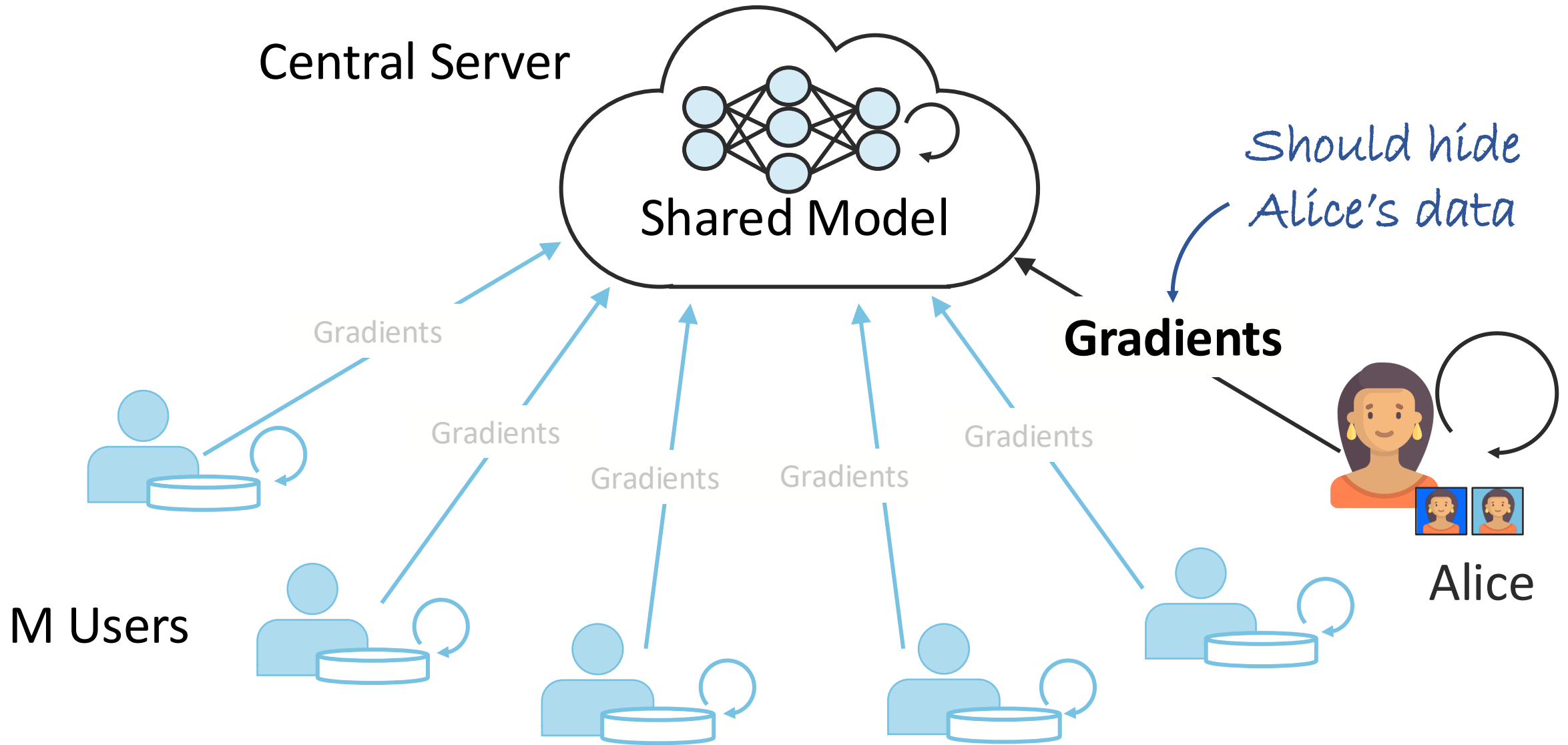


Centralized vs Federated Learning



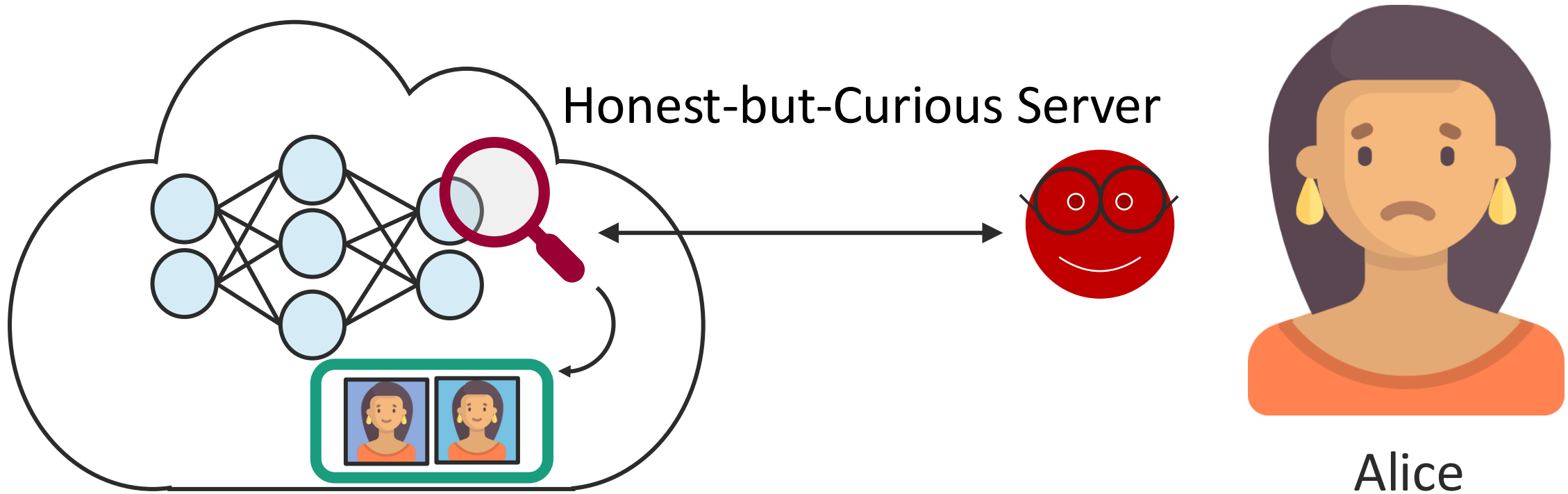


Alice's Privacy Relies Purely on the Gradients

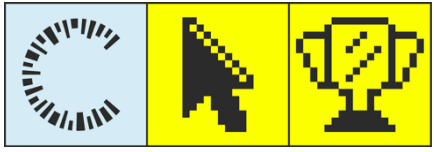




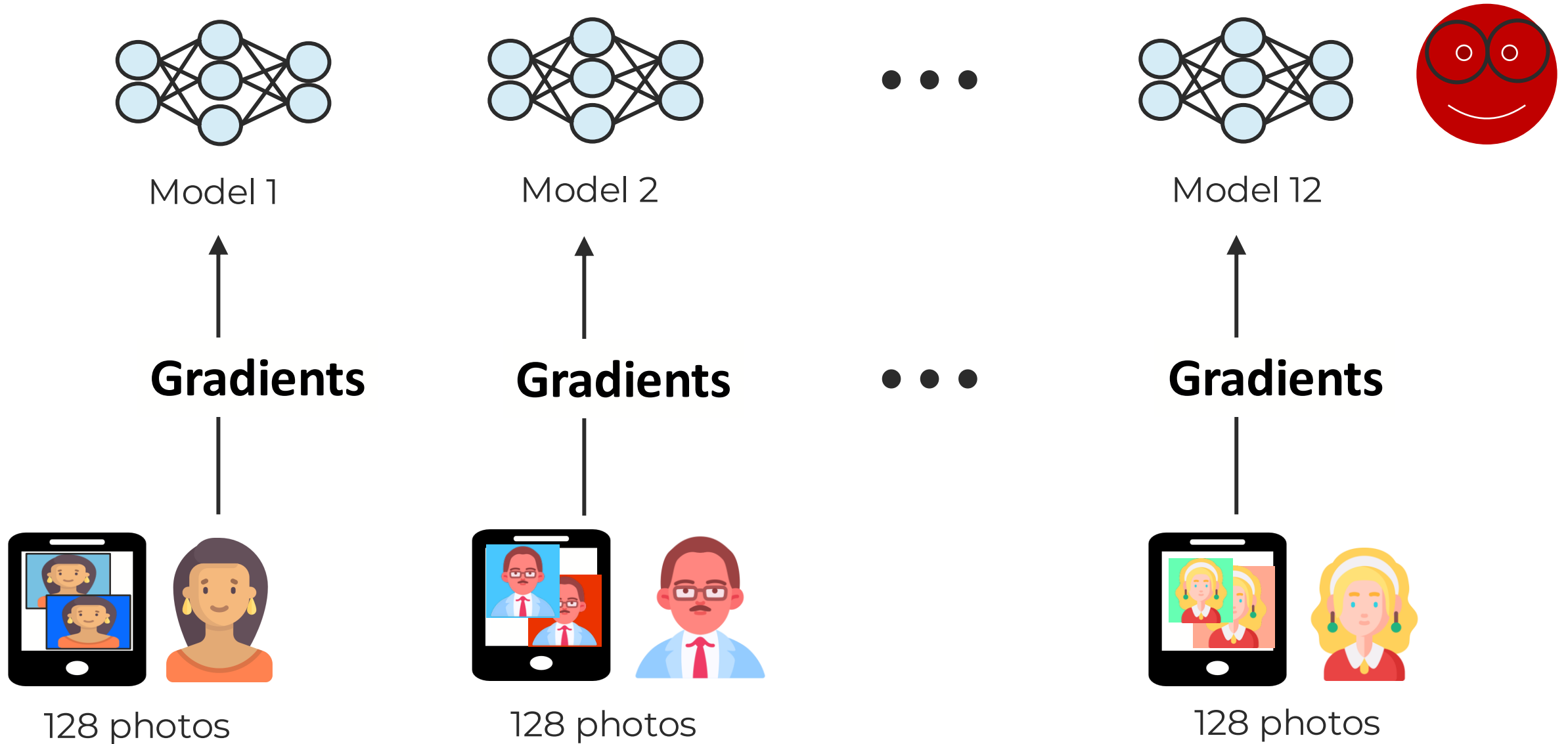
Task 3: Reconstruct Data from Gradients



You are a passive, honest-but-curious server who wants to extract a significant number of sensitive user photos from their returned gradients.



Task 3: Setup with Shared Models and Gradients





Task 3: Submission Format and Score

Submit a .pt file with **n=1536 reconstructed images**.

Provide 128 images each with dimension (3, 64, 64) per each of 12 models. Each pixel is float32 in [0,1]. Format of the submission:

```
{  
  "model1": Tensor(128, 3, 64, 64)  
  "model2": Tensor(128, 3, 64, 64)  
  ...  
}
```

$$\textbf{Score: } \frac{1}{n} \sum_{i=1}^n SSIM_i$$